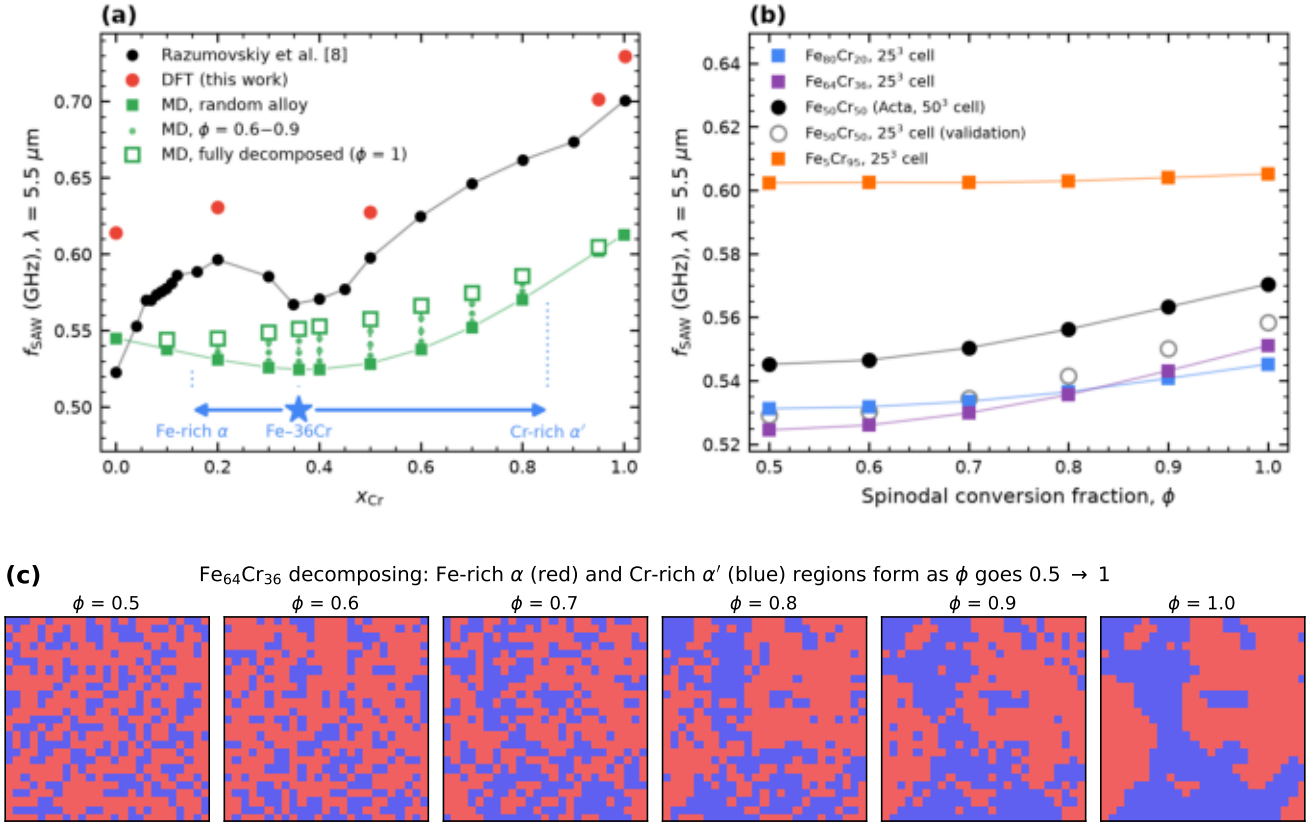

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, ϕ = 0.6–0.9
MD, fully decomposed (ϕ = 1)
Fe-rich α
Fe–36Cr
Cr-rich α'
f_{SAW} (GHz), λ = 5.5 μm
x_{Cr}
(b)
$Fe_{80}Cr_{20}$, 25^3 cell
$Fe_{64}Cr_{36}$, 25^3 cell
$Fe_{50}Cr_{50}$ (Acta, 50^3 cell)
$Fe_{50}Cr_{50}$, 25^3 cell (validation)
Fe_5Cr_{95}, 25^3 cell
f_{SAW} (GHz), λ = 5.5 μm
Spinodal conversion fraction, ϕ
(c)
$Fe_{64}Cr_{36}$ decomposing: Fe-rich α (red) and Cr-rich α' (blue) regions form as ϕ goes 0.5 → 1
ϕ = 0.5
ϕ = 0.6
ϕ = 0.7
ϕ = 0.8
ϕ = 0.9
ϕ = 1.0
one atomic (001) plane of a 25^3 bcc MD cell: red = Fe, blue = Cr